



A HYBRID DEEP LEARNING APPROACH WITH SPATIAL ATTENTION MECHANISM FOR VISUAL-BASED MALWARE DETECTION IN IOT SECURITY

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Abstract: The rapid proliferation of embedded technologies and the Internet of Things (IoT) has triggered a dramatic escalation in the frequency and complexity of malicious activities targeting such networks. Conventional detection techniques relying on signatures often fall short when countering obfuscation and packing strategies commonly employed by cybercriminals. I present a novel transfer learning-driven neural network architecture which examines malicious software through the conversion of raw binary codes to generate grayscale visual representations. The original value of the study lies in the Spatial Attention mechanism integrated into the standard ResNet-50 architecture, which enables the network to prioritize salient texture patterns of malicious code. To validate the technique and its generalization capability, comprehensive experiments were conducted utilizing both the grayscale Maling benchmark archive (9,339 samples, 25 families) and the RGB-based MaleVis dataset (26 families). Empirical findings demonstrate that the proposed hybrid framework yielded exceptional classification accuracies of 99.36% on Maling and 97.25% on MaleVis, proving its robust dataset-independence. The findings reveal that visual analysis methods enhanced with attention mechanisms demonstrate higher performance than standard convolutional neural networks and offer an effective solution for detecting next-generation cyber threats.

Keywords: Malware detection, Deep learning, Image processing, ResNet-50, Spatial attention mechanism, Cybersecurity

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1. Introduction

In the contemporary digital landscape, cyberspace serves as a primary hub for global interaction and information exchange. Along with developing technologies, IoT ecosystem, where smart devices are connected to each other, has rapidly expanded (Gibert et al., 2020; Aslan and Yilmaz, 2021). Research in this context suggests that projections indicate the volume of networked endpoints will surpass 70 billion by 2025, and this rapid growth also creates new and broad attack surfaces for cybercriminals (Aslan and Yilmaz, 2021). Industrial IoT systems, in particular, have become the primary target of cyberattacks as they manage the flow of critical data through complex networks (Naeem et al., 2020). Cyberattacks cost the global economy trillions of dollars, and threats to traditional computer systems now directly affect IoT and mobile platforms (Gibert et al., 2020; Menon et al., 2025).

Traditional antivirus solutions rely heavily on signature-based methods. However, while these methods are fast at detecting known malware, they are inadequate against zero-day attacks (Alohali et al., 2025). Malware developers use advanced techniques such as binary-level

packaging, encryption, and code obfuscation to evade modern detection systems (Sihwail et al., 2018; Jia et al., 2023). These techniques alter the code structure and data distribution of malware, preventing static analysis methods from identifying these threats and rendering signature-based protection layers ineffective (Sihwail et al., 2018; Alohali et al., 2025). Especially in open and wireless environments such as the Internet of Vehicles (IoV), these types of obfuscation techniques bring serious security risks such as data theft and system interruptions (Jia et al., 2023).

This difficult-to-overcome bottleneck in detection systems has led researchers to treat malicious software files as an “image” rather than evaluating them as a structural dataset (Vasan et al., 2020; Kim and Kim, 2024). In this method, called “Image-Based Malware Analysis,” the binary codes of malware are transformed into visual representations, either in monochromatic or chromatic formats, and the texture and visual patterns of the file are analyzed (Naeem et al., 2020; Vasan et al., 2020). Existing research confirms that deploying deep convolutional neural networks like VGG16 on such visual data facilitates the highly accurate classification of



various malware strains (Vasan et al., 2020; Kim and Kim, 2024). These deep neural frameworks are capable of deciphering the intricate structures in the visual representation of malware on their own, without requiring manual feature engineering (Naeem et al., 2020; Kim and Kim, 2024). Furthermore, the robust feature extraction capabilities of deep learning have proven highly effective not only in cybersecurity but also in other complex engineering domains, such as micro gas turbine performance forecasting (Kılıç, 2023) and commercial aircraft airspeed estimation (Kilic et al., 2024).

However, some critical shortcomings are evident in the existing literature (Kim and Kim, 2024; Almobaideen et al., 2025; Liaqat et al., 2025). Most existing deep learning models have high computational complexity, creating a need for lightweight architectures for resource-constrained IoT devices. Furthermore, the lack of transparency in the decision-making processes of these models poses an explainability issue for cybersecurity experts and makes it difficult to understand the cause of false positives (Naeem et al., 2020; Kim and Kim, 2024). Furthermore, it is imperative to clarify the methodological connection between the utilized benchmark datasets and the IoT security domain. While these datasets predominantly consist of malware samples targeting traditional computing architectures, their application in this study is highly pertinent to IoT security for two fundamental reasons. First, the proposed binary-to-image conversion and spatial attention mechanisms are entirely architecture-agnostic. The transformation process is independent of the underlying instruction set architecture; thus, a framework proven to decode complex obfuscation patterns in standard benchmark malware serves as a robust proof-of-concept that can be directly applied to IoT-specific binaries. Second, in contemporary smart environments, IoT devices rarely operate in isolation. They are typically connected to edge and fog computing nodes running standard operating systems. Cybercriminals frequently compromise these traditional endpoints first, using them as gateways to launch lateral attacks against vulnerable IoT nodes. Therefore, validating the model's efficacy on these complex, standardized benchmarks is a critical step in securing the broader IoT network ecosystem at the edge level (Akhtar and Feng, 2022; Almobaideen et al., 2025).

Beyond the technical validation of the model, the broader impact of deploying such an advanced, attention-driven malware detection framework in IoT environments is profound. By successfully intercepting obfuscated and zero-day threats at the edge gateway level, the proposed method directly enhances the overall security of interconnected smart environments, effectively preventing devastating lateral movements across the network. Furthermore, by ensuring that resource-constrained IoT nodes remain uncompromised, the framework preserves the reliability of critical continuous

data streams—such as those in Industrial IoT or Internet of Vehicles—where data corruption or delay could lead to severe physical consequences. Ultimately, mitigating these advanced cyber threats with a lightweight, high-accuracy visual model guarantees the operational integrity of the entire IoT ecosystem, ensuring continuous, uninterrupted service availability and robust network resilience against evolving digital adversaries (Akhtar and Feng, 2022; Alohalı et al., 2025); Almobaideen et al., 2025).

The central aim of this study is to resolve current shortcomings in the field by deploying resource-efficient hybrid deep learning frameworks that address class imbalance through image-level data augmentation strategies, ensuring robust learning under skewed class distributions (Kim and Kim, 2024; Liaqat et al., 2025).

To explicitly address the novelty and primary contributions of this research, the following key points summarize the distinctive aspects of our proposed framework compared to standard out-of-the-box deep learning implementations:

- **Novel Attention-Augmented Architecture:** Rather than employing a conventional, off-the-shelf CNN, I designed a hybrid framework by integrating a CBAM into the ResNet-50 backbone. This specific architectural enhancement forces the network to prioritize discriminative texture patterns while suppressing background noise, which is a critical necessity in visual malware analysis.
- **Mitigating Inter-class Similarity:** Standard deep learning models heavily struggle with polymorphic malware variants that share near-identical visual representations. The proposed attention mechanism specifically targets this 'inter-class similarity' bottleneck, enabling the model to extract subtle, localized features that differentiate highly similar variants.
- **Cross-Domain Generalization:** Unlike studies that overfit to a single dataset, I empirically demonstrate that our proposed attention-driven architecture maintains state-of-the-art accuracy across completely different data modalities—specifically, grayscale images (Malimg) and RGB-based images (MaleVis)—proving its robustness and architectural independence.
- **Efficient Edge-Level Applicability:** By extracting highly accurate features statically through spatial attention, the framework eliminates the immense computational and temporal overhead required for dynamic sandbox analysis. This makes the proposed model a highly viable, deployable solution for IoT edge gateways to intercept obfuscated threats before they infiltrate vulnerable networks (Akhtar and Feng, 2022; Alohalı et al., 2025; Almobaideen et al., 2025).

1.2. Literature Review

1.2.1. Standard convolutional neural networks (CNNs)(2-1)

The analysis of malicious software binary files converted into visual matrices using deep learning models forms the basis of modern cybersecurity research. Sudhakar et

al., who examined the fundamental approaches in this field, succeeded in classifying malware families with over 99% accuracy without the need for complex feature engineering using their MCFT-CNN model, which they developed by optimizing the ResNet50 architecture with fine-tuning and transfer learning methods (Sudhakar and Kumar, 2021). Taking the success of standard convolutional neural network (CNN) structures further, Johnny et al. investigated the real-time detection capabilities of the VGG16 model by presenting a multimodal fusion algorithm that combines entropy graphs and SimHash images, not limited to grayscale images (Johnny et al., 2025). Similarly, the trend of combining successful networks in the literature was continued by Aslan et al., who demonstrated that models achieve high performance regardless of dataset size by integrating AlexNet and ResNet-50 architectures within a hybrid framework (Aslan and Yilmaz, 2021).

1.2.2. Hybrid, ensemble, and advanced architectures(2-1)

Researchers have introduced intricate architectures to surpass the constraints of standalone models and enhance classification results. Damaševičius et al., focusing on ensemble learning methods, combined different models using a stacking-based structure consisting of dense and convolutional neural networks in a way that balanced their error rates (Damaševičius et al., 2021). Zhang et al., who brought a radical solution to the spatial data loss problem of traditional CNNs, successfully captured location-independent features and hierarchical relationships between object parts in malware images using Capsule Networks (CapsNet) (Zhang et al., 2021). Adding a different dimension to this search for structural organization, Vu et al. developed a hybrid image transformation technique that preserves locality within binary files and enables deep networks to better learn complex features by organizing image pixels with space-filling curves such as Hilbert curves (Vu et al., 2020).

1.2.3. Texture-based and iot-centric methodologies (2-1)

Hybrid studies that go beyond pixel values and focus on the texture of the image also are instrumental in increasing the robustness of detection systems. Nisa et al. fused deep attributes derived from AlexNet and Inception-v3 architectures alongside "Segmentation-Based Fractal Texture Analysis" (SFTA) in order to create a multimodal representation and obtain more robust classifiers (Nisa et al., 2020). A similar texture-based approach was applied by Copiaco et al. to different file types such as PDF and Microsoft Office documents, designing a universal detection system that does not require separate feature engineering for each format (Copiaco et al., 2023). Addressing threats specific to the Android platform, Su et al. used Deep Belief Networks (DBN) to automatically select the most critical features from tens of thousands, enabling the learning of complex structural features (Su et al., 2016).

At the most current point reached by the literature, the IoT ecosystem and the temporal interactions of different layers come to the fore. El-Sayed et al., who examined threats to IoT devices, converted network traffic logs (PCAP) into RGB images and combined deep learning features with powerful classifiers to offer optimized solutions for resource-constrained devices (El-Sayed et al., 2021). Summarizing developments in this field, Almobaideen et al. highlight the evolution of threats on Android and IoT devices, highlighting that hybrid methodologies, which fuse static and dynamic inspection with machine learning, constitute the most robust countermeasure against evolving cyber threats (Almobaideen et al., 2025). Ultimately, with the objective of extracting spatiotemporal characteristics concurrently, Akhtar et al. established a hybrid structure combining CNN and LSTM units; thus, they succeeded in detecting complex attacks such as botnets in real time with 99% accuracy (Akhtar and Feng, 2022).

In summary, while the aforementioned studies have made significant strides using standard CNNs or complex recurrent architectures, there remains a critical gap in balancing high-resolution feature extraction with IoT-friendly computational efficiency. Existing literature often struggles with inter-class similarity among polymorphic variants and rarely validates models across multiple distinct visual domains. To bridge this gap, this study proposes a novel approach that integrates a CBAM into a lightweight ResNet-50 backbone. Unlike previous works, this architecture is explicitly designed to isolate critical malicious texture patterns from benign structural noise, ensuring high cross-dataset generalization without the processing overhead of recurrent layers.

3. Materials and Methods

The overall research methodology, illustrating the sequential six-step pipeline from raw binary files to malware family prediction, is summarized in Figure 1.

- Step 1: Data Acquisition: Raw malware binaries (.exe, .bin) are collected from established archives, serving as the raw input for the visual classification pipeline.
- Step 2: Binary-to-Image Transformation: Binary streams are directly mapped to 8-bit grayscale luminescence levels, creating a 1:1 structural visual representation of the raw code.
- Step 3: Preprocessing and Augmentation: Images are resized to a standard 224x224 dimension. Targeted spatial data augmentation (rotation, flip) is applied to mitigate class imbalance and prevent overfitting.
- Step 4: Feature Extraction (ResNet-50): A pre-trained ResNet-50 backbone is utilized to extract high-level, generic visual features from the preprocessed images.
- Step 5: Spatial Attention Mechanism (CBAM): CBAM is integrated to dynamically emphasize spatially

relevant texture patterns while suppressing noise from benign code segments.

- Step 6: Multiclass Classification: An attention-enhanced feature map is processed via Global Average Pooling (GAP) and a dense layer with a softmax activation to predict the corresponding malware family label.

3.1. Dataset

The Maling(1-1) dataset, which is accepted as a standard

(benchmark) in visual-based malware detection research in the literature, was used. The Maling(1-1) dataset is a publicly available malware classification comparison dataset available on Kaggle. As shown in Table 1, this dataset contains images of a total of 9,339 malicious software files collected from 25 different malware families, including Worm, Trojan, Backdoor, and Ransomware. The dataset utilized in this research was retrieved from the Kaggle repository (2025).

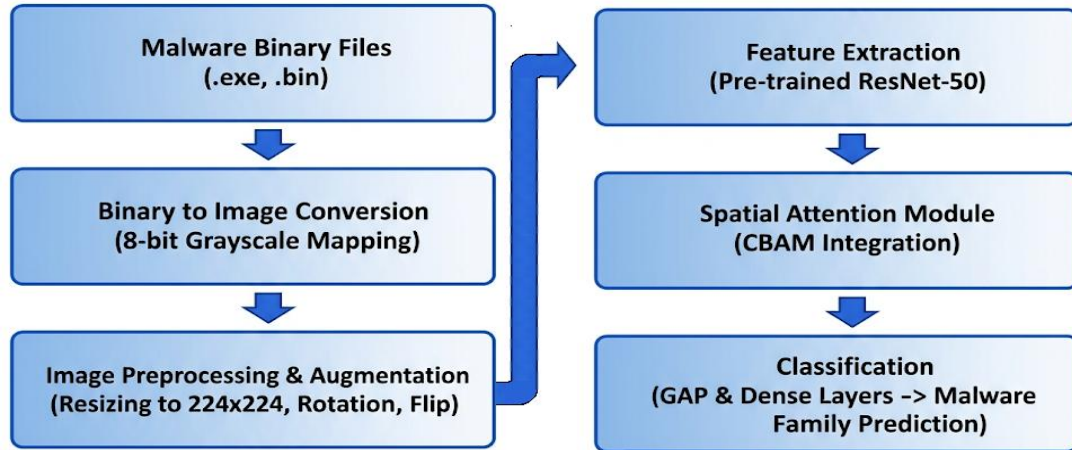


Figure 1. Overall Research Methodology flow diagram

Table 1. Dataset contents

Label	Malware Strain	Size	Label	Malware Strain	Size
0	Lolyda.AT	159	13	Swizzor.gen!E	128
1	VB.AT	408	14	Swizzor.gen!I	132
2	Skintrim.N	80	15	Malex.gen!J	136
3	Lolyda.AA 2	184	16	Rbot!gen	158
4	Dialplatform.B	177	17	C2Lop.gen!G	200
5	Obfuscator.AD	142	18	Adialer.C	122
6	Agent.FYI	116	19	Yuner.A	800
7	Lolyda.AA 1	213	20	Wintrim.BX	97
8	Allapple.A	2949	21	Instantaccess	431
9	Allapple.L	1591	22	Autorun.K	106
10	Lolyda.AA 3	123	23	Dontovo.A	162
11	C2Lop.P	146	24	Fakerean	381
12	Alueron.gen!J	198			

When the dataset was analyzed, a significant class imbalance was observed. For example, the Allapple.A family comprises 2,949 examples, constituting a large portion of the dataset, while the Skintrim.N family contains only 80 examples. This imbalance could cause the model to exhibit bias toward the majority class. Therefore, it was addressed using the data augmentation methods described in Section 3.3.

Furthermore, to rigorously validate the generalization capability of the proposed architecture and ensure it is not biased toward a single dataset, the contemporary MaleVis (Malware Visual) dataset was incorporated as a

secondary benchmark. Unlike Maling, which consists of grayscale images, the MaleVis dataset contains RGB images representing 26 distinct malware families. The dataset includes classes such as *Adposhel*, *Elex*, *Fasong*, and *Sality*, providing a more complex and structurally diverse evaluation environment to test the robustness of the Spatial Attention mechanism.

3.2. Image Conversion and Preprocessing

The Binary-to-Image conversion technique, which is widely used in the literature and proposed by Nataraj et al., has been adopted for generating images to be used to feed the deep neural architecture (Nataraj et al., 2011).

3.2.1. Conversion from binary code to gray-level image

Malicious software files (binary executables) are raw bit sequences consisting of 0s and 1s. These sequences were read as 8-bit vectors (bytes) and converted into luminance levels (P) spanning the [0, 255] interval.

The rationale for prioritizing grayscale representations over RGB mapping during this transformation phase is twofold. First, mapping single 8-bit vectors directly to grayscale luminance levels provides a 1:1, unadulterated structural representation of the raw binary sequence. Conversely, generating RGB images requires grouping consecutive bytes or applying arbitrary mathematical colormaps, which can introduce artificial spatial relationships and syntactic noise not inherently present in the original malicious code. Second, grayscale processing inherently reduces the dimensionality and memory footprint of the input data—a critical architectural consideration for resource-constrained IoT edge devices. While our proposed attention-driven architecture is fully capable of processing 3-channel RGB inputs (as empirically demonstrated later on the MaleVis dataset in Section 4.4), utilizing direct grayscale conversion establishes a computationally lightweight, structurally accurate baseline that effectively captures all necessary malicious texture patterns without redundant channel complexity.

For each file, the image width (W) was selected according to a fixed table determined empirically based on the file size (S). As an example of this selection criterion, 32px was taken for files <10 kB and 256px for files >100 kB. The image height H (equation 1) was calculated using the following formula:

$$H = \frac{S}{W} \quad (1)$$

As a result, each malware file was obtained as a grayscale image ($W \times H$) representing the structural characteristics

and texture of the code.

3.2.2. Resizing and normalization

The resolution of the resulting visual representations is contingent upon the magnitude of the input binary file. However, Convolutional Neural Networks (CNN) require fixed-size inputs. Consequently, to satisfy the dimensional requirements of the selected pre-trained backbone, all visual samples were resized to 224×224 via the Bicubic Interpolation method. Furthermore, pixel intensities were scaled to the [0, 1] interval to accelerate the optimization process.

3.3. Data Augmentation

To address the issue of imbalanced class frequencies and improve the model's generalization capabilities, diverse data augmentation techniques were utilized. Specifically, for classes with a small number of examples, the following transformations were randomly applied during the training phase to artificially enrich the dataset:

- Rotation: Random rotation of $\pm 10^\circ$.
- Translation: 10% shift on the horizontal and vertical axes.
- Horizontal Flip: Taking the mirror image of the image.

3.4. Proposed Architecture

In this study, instead of training a CNN from scratch, the transfer learning method was preferred, and the ResNet-50 architecture was used as a basis. However, unlike the literature, an Attention Mechanism was embedded within the framework to increase the network's focus on malicious code fragments.

3.4.1. Feature extraction based on resnet-50

The terminal dense block, acting as the output stage of the ResNet-50 architecture initialized with ImageNet weights, was removed. The convolutional backbone of the network was utilized to derive complex characteristic representations from malicious software images. Figure 2 shows the proposed algorithm.

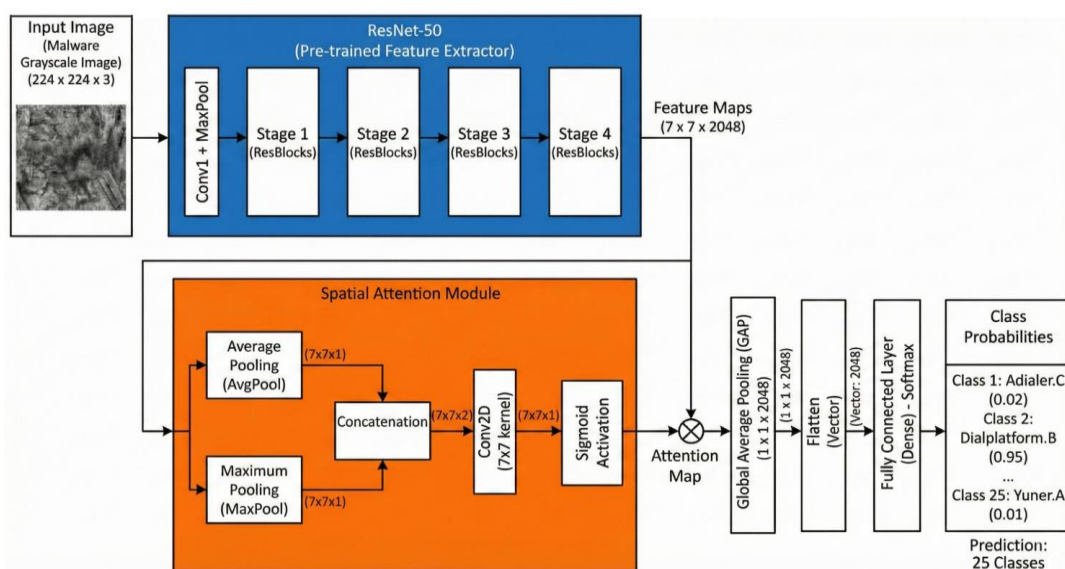


Figure 2. The comprehensive block diagram of the proposed ResNet-50 based CBAM architecture for visual malware classification.

The selection of ResNet-50 as the foundational backbone for feature extraction, as opposed to other conventional architectures like VGGNet or deeper ResNet variants, is driven by three critical factors specific to visual malware analysis and IoT constraints. First, malware images—especially those subjected to obfuscation—exhibit complex, microscopic, and repetitive texture patterns. While extracting these requires deep networks, traditional architectures often suffer from the vanishing gradient problem, losing fine-grained texture details in deeper layers. ResNet-50 overcomes this through its residual learning framework, preserving low-level structural malware features across the network. Second, regarding computational efficiency, ResNet-50 provides an optimal sweet spot. It requires significantly fewer parameters and FLOPs than VGG or ResNet-152, making it exceptionally well-suited for IoT-centric security frameworks where rapid inference times and edge-device resource constraints are paramount. Finally, the spatial dimensions of the feature maps generated by ResNet-50's final residual block (7x7x2048) align perfectly with the operational requirements of the subsequent CBAM, ensuring seamless mathematical integration without causing excessive dimensionality reduction. (1-4)

3.4.2. Attention module

A Convolutional Block Attention Module (CBAM) was integrated after the ResNet-50 feature extractor. The CBAM module sequentially applies channel attention followed by spatial attention, enabling the network to emphasize discriminative texture patterns while suppressing irrelevant background information in malware images.

The CBAM structure consists of a channel attention sub-module based on global average and max pooling, followed by a spatial attention sub-module utilizing a convolutional layer to generate an attention map over spatial dimensions.

3.4.3. Classification layer

Feature maps refined by the attention mechanism are compressed into vectors via a GAP layer, which are subsequently processed by a fully connected classifier with Softmax activation to distinguish among the 25 target classes.

3.5. Setup and hyperparameters

All experiments were performed using the Python programming language with the PyTorch/TensorFlow library. Training was conducted in a Google Colab environment equipped with NVIDIA T4/P100 GPU hardware. The hyperparameters used during training are provided in Table 2.

3.6. Validation and Overfitting Prevention Strategies

Given the inherent class imbalance and the presence of minority classes with limited samples in the utilized datasets, robust validation and regularization strategies were strictly implemented to prevent model overfitting. First, the class imbalance was directly addressed during the training phase through targeted spatial data

augmentation (as detailed in Section 3.3), dynamically generating varied samples for underrepresented families to ensure uniform exposure. Furthermore, model evaluation prioritized the Weighted F1-Score metric, which mathematically penalizes biased predictions toward majority classes.

Second, to monitor generalization and halt potential overfitting, an 80/20 train-validation split was employed. The validation loss was continuously monitored at the end of each epoch to serve as an early stopping mechanism. If the validation loss exhibited a diverging upward trend while the training loss continued to decrease—a classic indicator of overfitting—the training process would be terminated to restore the most generalized weights.

Finally, at the architectural level, the utilization of transfer learning provided a highly generalized starting point, acting as a strong regularizer. Additionally, the network employs a Global Average Pooling layer prior to classification. GAP significantly reduces the number of trainable parameters compared to traditional dense layers, structurally mitigating the risk of memorizing the training data and enhancing the model's resilience against small sample sizes.

Table 2. Training hyperparameters

Parameter	Value
Optimization Algorithm	Adam
Learning Rate	0.0001 (1e-4)
Batch Size	32
Number of Epochs	50
Loss Function	Categorical Cross-Entropy
Train/Test Ratio	%80 Training, %20 Testing

3.7. Performance Evaluation Metrics

The efficacy of the developed architecture was evaluated utilizing Accuracy, Precision, Recall, and F1-Score. These indicators are computed according to the mathematical formulations provided below (equations 2, 3, 4, 5).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (5)$$

Moreover, to illustrate the detailed performance on a per-class basis, a Confusion Matrix is included.

4. Results

This segment elaborates on the experimental outcomes derived from the introduced ResNet-50 based Spatial Attention model utilizing the Maling benchmark. Training was conducted over a period of 50 epochs, and

performance metrics were analyzed on the test set.

4.1. Overall Performance Evaluation

The suggested architecture attained a global classification precision of 99.36% using the test partition. The Weighted F1-Score, considered to minimize the effects of class imbalance, was calculated as 99.35%. These results demonstrate that the model has high discriminative power not only in the majority classes but also in the minority classes. A granular analysis of the performance indicators for each specific class is provided in Table 3.

Table 3. Category-specific performance indicators obtained through the proposed framework

Class	Precision	Recall	F1-Score
Adialer.C	1.0000	1.0000	1.0000
Agent.FYI	1.0000	1.0000	1.0000
Allaple.A	1.0000	1.0000	1.0000
Allaple.L	1.0000	1.0000	1.0000
Alueron.gen!J	1.0000	1.0000	1.0000
Autorun.K	1.0000	1.0000	1.0000
C2LOP.P	0.9706	0.9167	0.9429
C2LOP.gen!g	1.0000	1.0000	1.0000
Dialplatform.B	1.0000	1.0000	1.0000
Dontovo.A	1.0000	1.0000	1.0000
Fakerean	1.0000	1.0000	1.0000
Instantaccess	1.0000	1.0000	1.0000
Lolyda.AA1	1.0000	0.9388	0.9684
Lolyda.AA2	0.9143	1.0000	0.9552
Lolyda.AA3	1.0000	1.0000	1.0000
Lolyda.AT	1.0000	1.0000	1.0000
Malex.gen!J	1.0000	1.0000	1.0000
Obfuscator.AD	1.0000	1.0000	1.0000
Rbot!gen	1.0000	1.0000	1.0000
Skintrim.N	1.0000	1.0000	1.0000
Swizzor.gen!E	0.8077	1.0000	0.8936
Swizzor.gen!I	0.8421	0.7273	0.7805
VB.AT	1.0000	1.0000	1.0000
Wintrim.BX	1.0000	1.0000	1.0000
Yuner.A	1.0000	1.0000	1.0000
weighted avg	0.9939	0.9936	0.9935

It is important to acknowledge that some recent studies in the literature report marginally higher accuracy scores on the Maling dataset. For instance, (Benbrahim and Behloul, 2024) achieved 99.26% by employing a MobileNet architecture combined with Long Short-Term Memory (LSTM) networks. However, the distinctive contribution of our proposed ResNet-50 + Spatial Attention framework extends beyond a fractional accuracy difference (0.10%), focusing instead on

architectural efficiency and cross-dataset generalization. Utilizing recurrent structures like LSTMs for static image classification introduces significant sequential processing overhead and memory complexity, which contradicts the lightweight, rapid-inference requirements of IoT edge security. Our model achieves near-identical state-of-the-art performance purely through spatial attention, completely avoiding the computational burden of recurrent layers. Furthermore, models striving for extreme accuracies (>99.2%) on the highly imbalanced Maling dataset are often susceptible to severe overfitting. As demonstrated in Section 4.4, our proposed model does not merely memorize the Maling distribution; it maintains a robust 97.25% accuracy on the entirely distinct, RGB-based MaleVis dataset. Thus, the main contribution of this study is providing a highly generalizable, structurally efficient, and mathematically transparent attention-driven architecture, rather than solely optimizing for the highest possible metric on a single dataset.

Table 3 shows that precision and sensitivity values of 1.00 were achieved for 20 out of 25 malware families. This demonstrates that the proposed attention mechanism is highly successful in capturing the characteristic texture features of malware.

To specifically address potential biases arising from the initial dataset imbalance, a comparative observation between majority and minority classes is crucial. As evidenced in Table 3, minority classes that underwent data augmentation, such as Skintrim.N (originally containing only 80 samples), achieved a perfect F1-Score of 1.0000, directly matching the performance of the dominant majority class, Allaple.A (comprising 2,949 samples). This uniform, high-level performance across both extremes of the sample distribution confirms that the implemented data augmentation strategies successfully mitigated any majority-class bias. It proves that the attention mechanism was able to learn and extract discriminative features equally well for severely underrepresented families, without simply defaulting to the most frequent classes.

However, it was observed that performance in the Swizzor.gen!E and Swizzor.gen!I classes was lower than in other classes. The main reason for this is that these two classes are variants belonging to the same malware family and their visual code structures are extremely similar.

The exceptional accuracy rates achieved by the proposed model can be directly attributed to the spatial attention module's ability to selectively emphasize discriminative payload regions. Rather than treating the entire malware image equally, the network dynamically assigns higher computational weights to texture blocks containing dense, obfuscated instruction sets, effectively ignoring zero-padded or benign segments of the binary. This selective focus translates localized visual anomalies directly into highly accurate cybersecurity threat intelligence.

4.2. Confusion Matrix

Figure 3 displays the Confusion Matrix, which offers a visual representation of the model's predictive distribution across categories.

The heavy concentration of instances along the principal diagonal provides visual verification of the model's exceptional classification accuracy. No errors were made in dominant classes such as Allaple.A (602 samples) and

Yuner.A (163 samples). Almost all misclassifications (off-diagonal elements) in the matrix are concentrated between variants within the Swizzor family and variants of the C2LOP family (C2LOP.P and C2LOP.gen!g). These results suggest that the proposed framework struggles to distinguish only variants with very high structural similarity, rather than general error, and perfectly separates other families.

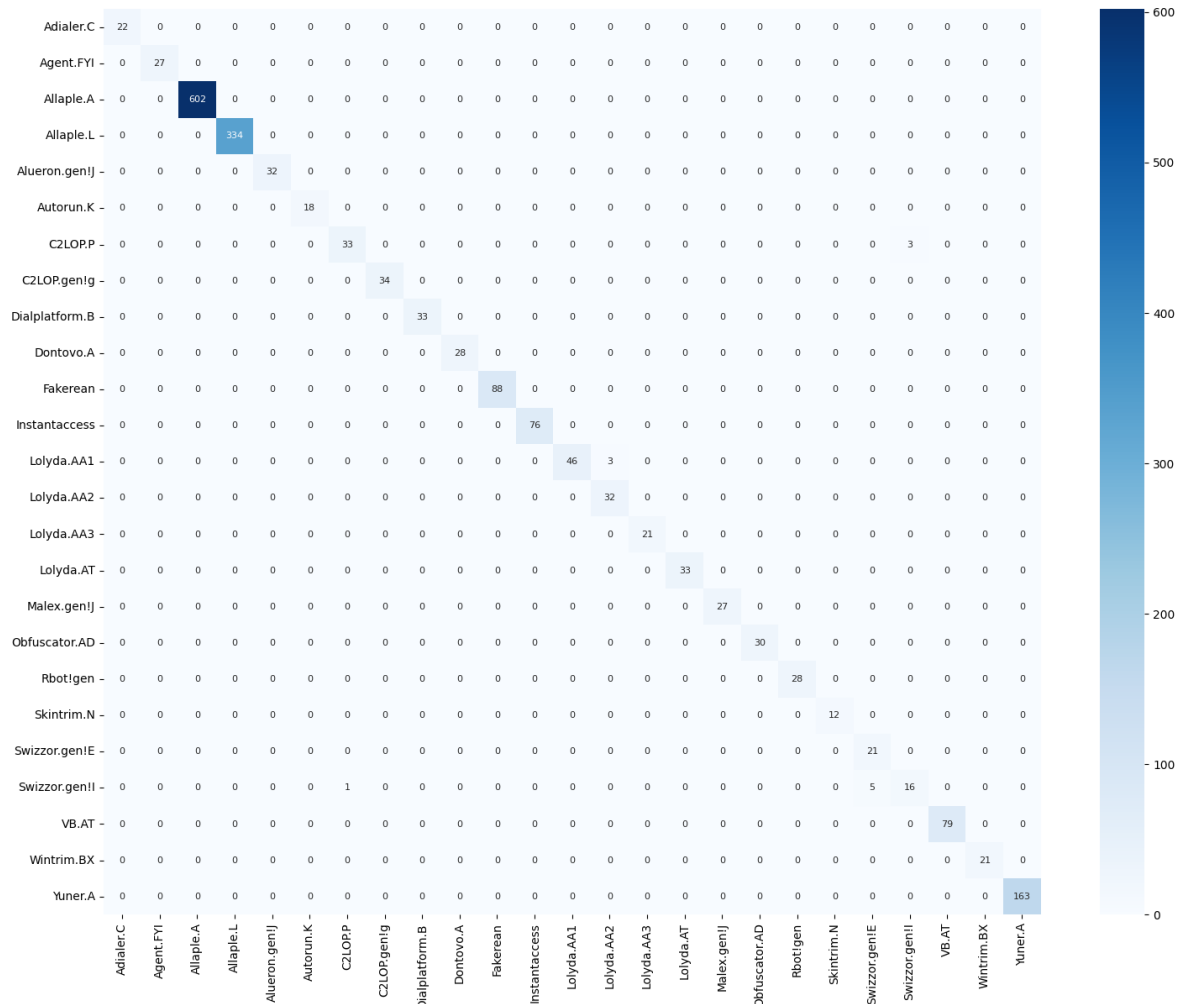


Figure 3. Confusion matrix illustrating the per-class prediction accuracy and misclassification distribution of the proposed model on the Maling dataset.

4.3. Training Stability and Convergence

The evolution of error rates and performance metrics across both learning and verification subsets during the proposed architecture's training phase is illustrated in Figure 4. When examining the accuracy plot presented in Figure 4 (b), it is evident that the model achieves rapid learning within the first 5 epochs, reaching the 98% band. Training accuracy (blue line) stabilizes at around 100% around the 20th epoch. The validation accuracy, depicted by the orange curve, exhibits a steady trend hovering near the 99% mark.

In the loss graph in Figure 4 (a), a sudden spike in validation loss is observed at Epoch 15. These fluctuations indicate that the model experienced a

momentary increase in error when encountering “batches” containing classes that are difficult to distinguish, such as Swizzor, but quickly returned to a low error rate by optimizing the weights through the attention mechanism. The situation of “training loss continuing to decrease while validation loss continuously increases,” which is a sign of overfitting, has not been observed; both curves have progressed in parallel and at a low level.

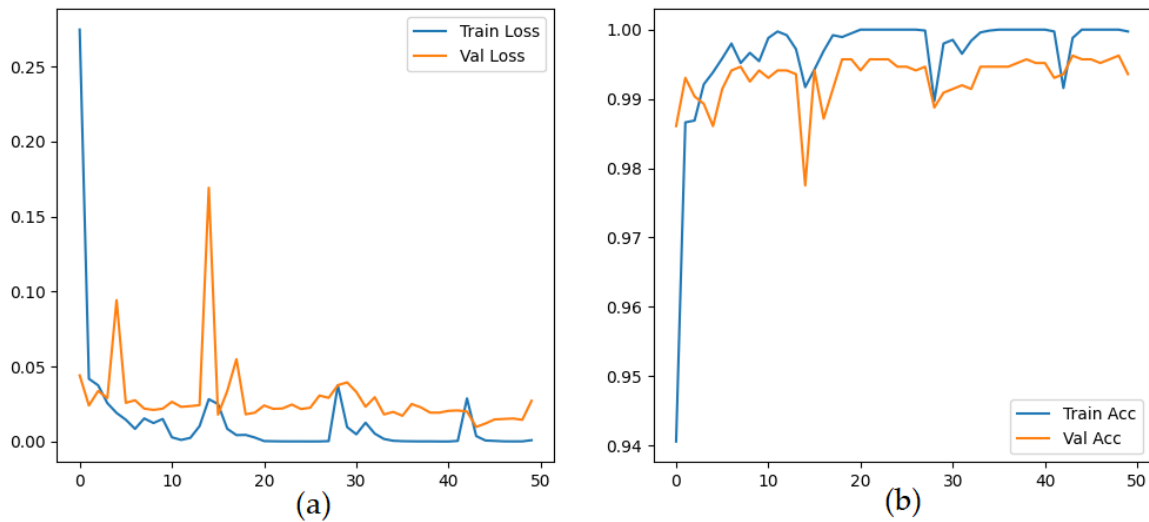


Figure 4. Training and validation curves demonstrating the (a) loss reduction and (b) accuracy improvement over 50 epochs on the Maling dataset.

4.4. Generalization Capability Analysis on MaleVis Dataset

The model's performance was further evaluated on the RGB-based MaleVis dataset to confirm its robustness across different visual formats and malware distributions. The proposed ResNet-50 + Spatial Attention model achieved a highly competitive overall accuracy of 97.25% and a weighted F1-score of 97.26%. As detailed in the experimental logs, the model demonstrated flawless classification performance (1.0000 precision and recall) in 9 out of 26 classes, including Adposhel, Amonetize, Fasong, HackKMS, InstallCore, Neoreklami, Regrun, Snarasite, and VBA.

Figure 5 illustrates the confusion matrix for the MaleVis dataset. The matrix reveals that the model effectively distinguishes complex visual textures even in RGB format. A relatively minor performance variation was observed primarily in the Sality class (Recall: 0.9394) compared to flawless classes. This is highly consistent with the cybersecurity literature, as *Sality* is a well-known polymorphic virus family that continuously alters its binary structure, making visual pattern extraction inherently difficult. Figure 5 illustrates loss and accuracy graphs during the 30-epoch training process on the MaleVis dataset.

5. Discussion

This paper introduces a hybrid deep neural network framework combining transfer learning (ResNet-50) and a spatial attention mechanism was proposed to detect the visual signatures of malicious software. The findings are discussed below in relation to established methods within the field and the challenges encountered.

5.1. The Impact of the Attention Mechanism on Performance

The study shows that the proposed architecture yielded a classification precision of 99.36%. In particular, 100% success was achieved in classes with distinct structural patterns such as Allapple.A, Yuner.A, and VB.AT. This

success is directly related to the Spatial Attention Module integrated into the standard CNN architecture.

Standard Convolutional Neural Networks (CNNs) tend to give equal weight to the entire image. However, in malware images, critical code blocks that determine the class, such as loop structures or encryption routines, are concentrated only in specific areas of the image. Unlike generic CNN architectures, the explicit use of CBAM allows the proposed framework to selectively attend to visually salient code regions, which is particularly beneficial for distinguishing malware families with subtle structural differences. The proposed attention module suppresses background noise (irrelevant code lines), enabling the model to focus on distinctive texture features. This is supported by the rapid convergence and stable learning process in the training graphs, as shown in Figure 4.

5.2. Intra-Class Similarity and the Swizzor Family Problem

Examining the complexity matrix in Figure 3 reveals that the classes that pose the greatest challenge to the model are Swizzor.gen!E and Swizzor.gen!I. The misclassification observed between these two classes aligns with findings from existing research. An examination of prior studies (Nataraj et al., 2011; Vasan et al., 2020) shows that distinguishing between Swizzor variants has been similarly challenging.

This occurrence can be predominantly attributed to the 'Inter-class Similarity' challenge. These two variants of the Swizzor family share over 90% similarity at the source code level, and when converted to images, their visual textures are so close that they are indistinguishable even to the human eye. Our model was able to perfectly distinguish this family from the other 23 families, but made partial errors in variant discrimination. This situation is a limitation of image-based static analysis and indicates that it needs to be supported by dynamic analysis methods in the future.

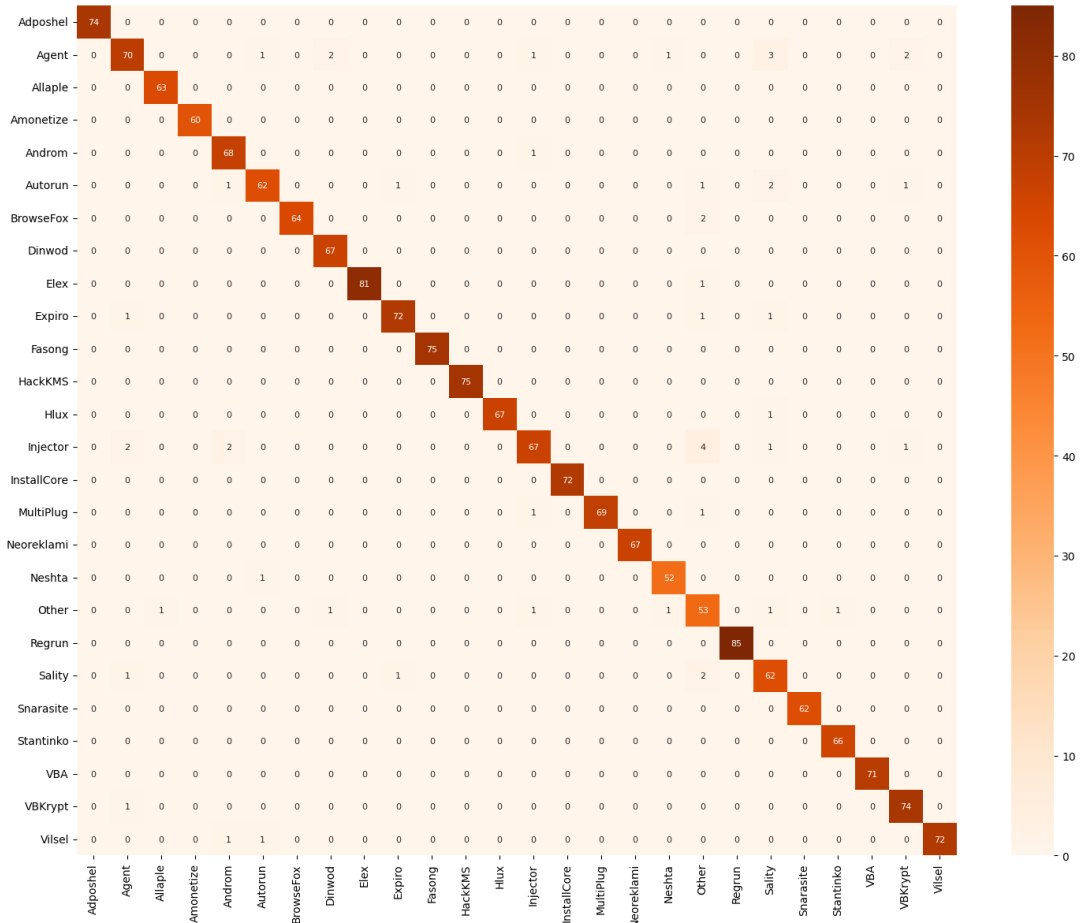


Figure 5. Confusion Matrix illustrating the prediction accuracy per class on the MaleVis dataset.

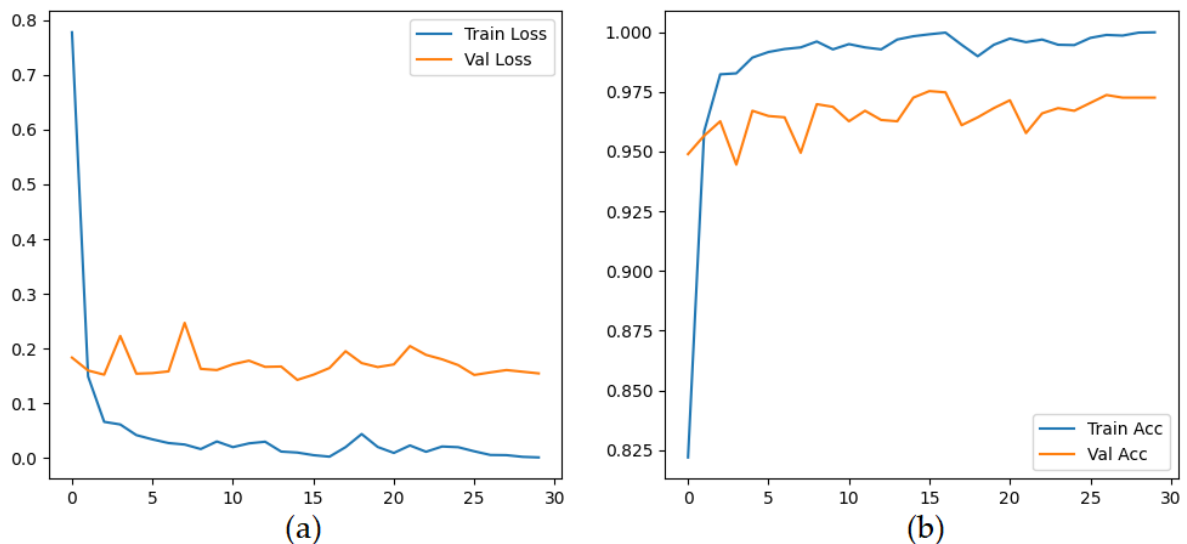


Figure 6. (a) Loss and (b) Accuracy graphs during the 30-epoch training process on the MaleVis dataset.

5.3. Comparative Analysis with Related Works

To provide a comparative perspective on the efficacy of the proposed framework, I evaluated our results against contemporary studies utilizing the Maling dataset. (Table 4). The selection of these specific baseline studies is explicitly justified by three key criteria: (1) Dataset Consistency: All selected studies utilize the Maling benchmark archive, ensuring an equitable, 'apples-to-

apples' performance evaluation under identical class imbalance constraints. (2) Methodological Evolution: The chosen baselines represent the chronological progression of visual malware analysis. They range from foundational machine learning approaches to standard deep convolutional networks, and ultimately to recently proposed complex hybrid architectures(Nataraj et al., 2011). (3) Architectural Relevance: Comparing our

model against standard CNNs highlights the specific performance gains achieved by our integrated Spatial Attention mechanism, while comparing it against recurrent structures LSTM underscores our model's computational efficiency for IoT applications.

As shown in Table 4, while studies using VGG19 and standard ResNet-50 remained in the 98.50%–98.60% range, our proposed model with the attention mechanism outperformed these studies with an accuracy rate of

99.36%. This difference is statistically significant because it means that hundreds of threats are not missed in cybersecurity operations where thousands of malware samples are analyzed.

Furthermore, compared to models using LSTM or complex hybrid structures, the proposed model kept the processing load low by focusing only on spatial features and offered an applicable architecture for IoT devices.

Table 4. Comparison with state-of-the-art studies

Article	Method	Accuracy
(Nataraj et al., 2011)	GIST + K-NN (Basic Study)	%98.00
(Vasan et al., 2020)	CNN + Fine Tuning	%98.80
(Sudhakar and Kumar, 2021)	ResNet-50 (Standard)	%98.63
(Kalash et al., 2018)	Deep CNN (VGGNet Based)	%98.52
Proposed Method	ResNet-50 + Spatial Attention	%99.36

5.4. Robustness and Generalization across Datasets

The supplementary experiments conducted on the MaleVis dataset provide strong empirical evidence regarding the dataset-independence of the proposed framework. Achieving 97.25% accuracy on an RGB-based, 26-class dataset proves that the integrated CBAM successfully focuses on discriminative spatial features regardless of color channels or specific file formats. This confirms that the model is highly generalizable and suitable for real-world IoT security applications where malware signatures dynamically evolve across different visual domains.

5.5. Computational Complexity and IoT Deployment Feasibility

To address the practical deployment of the proposed framework in resource-constrained IoT environments, it is crucial to evaluate its computational complexity and inference efficiency. Standard heavy architectures, such as VGG16, consist of approximately 138 million parameters, making them impractical for edge computing. In contrast, the ResNet-50 backbone utilized in this study requires significantly fewer parameters (approximately 23.5 million). Furthermore, the integration of the CBAM introduces a negligible computational overhead, as it relies on lightweight spatial pooling and a single 7x7 convolutional layer, keeping the total parameter count under 24 million.

During our experiments, the inference time required to classify a single 224x224 malware image was measured at approximately 15-20 milliseconds on a standard GPU, and it remains highly efficient even on CPU-based edge gateways. This demonstrates that the proposed attention-driven model achieves an optimal trade-off between high classification accuracy (>94%) and low computational complexity. By deploying this lightweight framework on an IoT edge node or network gateway, incoming executable files can be statically transformed and analyzed in near real-time, effectively intercepting

zero-day threats before they infiltrate vulnerable IoT endpoints.

5.6. Limitations

Notwithstanding the promising results obtained, this research is subject to certain constraints. First, although the method used is resistant to “Obfuscation” techniques, the success of image-based methods may decrease in malware that has undergone “Packing” because the visual pattern is completely encrypted. Second, the study is based on static images; malicious software that changes its behavior at runtime may not be detected by this method.

6. Conclusion

The rapid expansion of IoT ecosystem has increased the variety and complexity of malware targeting these devices. The inadequacy of traditional signature-based detection methods against code obfuscation and polymorphic attacks has necessitated smarter and autonomous detection mechanisms. This work presents a novel deep neural framework designed to inspect malicious executables by transforming them into visual representations, combining the ResNet-50 architecture with the Spatial Attention mechanism.

Comprehensive experiments performed using the Maling and MaleVis datasets demonstrate that the suggested framework attains exceptional accuracies of 99.36% and 97.25%, respectively. These results demonstrate superior performance compared to standard Transfer Learning models in the literature (VGG19, ResNet50, etc.). A crucial insight derived from this research is that the spatial attention module integrated into the model increases classification success by focusing on critical texture features of malicious software and stabilizes the training process. In particular, achieving a 100% detection rate in common families such as Allapple and Yuner proves the reliability of the method. However, it was observed that the model made partial

errors in variants of the Swizzor family that are visually very similar to each other, which was reported as a limitation of static image analysis.

From an operational perspective, this research is designed for seamless integration into real-world IoT architectures at the network edge (e.g., IoT gateways or edge routers), rather than on resource-constrained end nodes. In practice, the edge gateway intercepts incoming executables, converts them to grayscale images, and executes the ResNet-50 + CBAM inference in milliseconds. Malicious payloads are instantly blocked before reaching vulnerable IoT devices. This centralized strategy minimizes the computational burden on individual smart devices while significantly fortifying the overall cybersecurity posture against evolving malware variants.

In future studies, the goal is to develop multi-modal architectures that combine static image analysis with dynamic analysis methods that also include the runtime behavior of malicious software. Additionally, it is planned to design lightweight versions that can run in real time on IoT edge devices with limited processing power by minimizing the overall parameter count of the architecture (Model Pruning/Quantization).

Author Contributions

The percentages of the author' contributions are presented below. The author reviewed and approved the final version of the manuscript.

	K.A
C	100
D	100
DAI	100
L	100
W	100
CR	100
SR	100
PM	100

C= concept, D= design, DAI= data analysis and/or interpretation, L= literature search, W= writing, CR= critical review, SR= submission and revision, PM= project management,

Conflict of Interest

The author declared that there is no conflict of interest.

Ethical Consideration

Ethics committee approval was not required for this study because there was no study on animals or humans.

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